



### **NORMATIVE**

$T = I \quad \det = +1 \quad (x, y) \rightarrow (x, y)$

"This is the standard."

**RFC 2119 MUST / SHALL**

## **DNS Tool**

Communication Standards — Clarity & Vision Quality Gate

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**PROJECT** [dnstool.it-help.tech](https://dnstool.it-help.tech) **SOURCE** [github.com/IT-Help-San-Diego/dns-tool-intel](https://github.com/IT-Help-San-Diego/dns-tool-intel)

**VERSION** 26.46.12 · **LICENSE** BUSL-1.1

Companion to [Founder's Manifesto](#) and [Philosophical Foundations](#)

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## **The Owl Semaphore System**

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The Owl of Athena encodes document classification as semantic state through orientation and color. Each designation is a distance-preserving isometry drawn from the orthogonal group  $O(2)$ . The system uses four transforms — the Klein four-group  $V_4$  — to classify every publication in the DNS Tool corpus.

|                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                             |
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|  <p><b>NORMATIVE</b></p> $T = I \quad \det = +1$ $(x, y) \rightarrow (x, y)$ <p>"This is the standard."</p> <p>RFC 2119 MUST / SHALL</p> |  <p><b>NON-NORMATIVE</b></p> $T = \sigma_v \quad \det = -1$ $(x, y) \rightarrow (-x, y)$ <p>"This reflects the standard."</p> <p>Informative / Advisory (NOTE)</p> |  <p><b>CRITICAL</b></p> $T = C_2 \quad \det = +1$ $(x, y) \rightarrow (-x, -y)$ <p>"This inverts the standard."</p> <p>RFC 2119 MUST NOT / SHALL NOT</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Redundant encoding:** Color, orientation, and context label provide three independent channels. The semaphore remains distinguishable at small render sizes and in degraded single-channel contexts. Each transform is a distance-preserving isometry in  $O(2)$ . **Standards mapping:** Normative = RFC 2119 MUST/SHALL. Non-normative = Informative/NOTE. Critical = RFC 2119 MUST NOT/SHALL NOT; CVE/Security advisory. Metacognitive = Observer audit/Frame inversion.

## 1. Dual-Gate Requirement

Every piece of user-facing text must pass **both** gates simultaneously. Passing one gate but failing the other is a failure.

### Gate 1: Clarity

The words chosen must be clear, communicative, and logical to the reader. Text should educate, not obscure.

| REQUIREMENT           | LEVEL         | MEASUREMENT                                             |
|-----------------------|---------------|---------------------------------------------------------|
| Jargon with first use | <b>MUST</b>   | Every acronym expanded on first use per page, or linked |
| Active voice          | <b>SHOULD</b> | Prefer "The SPF record authorizes..." over passive      |

|                          |        |                                                               |
|--------------------------|--------|---------------------------------------------------------------|
| Sentence length          | SHOULD | Body copy sentences should not exceed 35 words                |
| One idea per paragraph   | SHOULD | Each paragraph should advance one concept                     |
| Action-oriented findings | MUST   | State what was observed, why it matters, what to do           |
| RFC citation             | MUST   | Claims about protocol behavior must cite specific RFC section |
| Transformative language  | SHOULD | Prefer language that teaches rather than restates             |

## Gate 2: Vision

The visual presentation must not cause strain or impede comprehension. The reader's eyes matter.

| REQUIREMENT                  | LEVEL  | MEASUREMENT                                                |
|------------------------------|--------|------------------------------------------------------------|
| Minimum font size            | MUST   | No rendered text below 0.75rem (12px)                      |
| Body text minimum            | SHOULD | Primary reading text at least 0.875rem (14px)              |
| Contrast ratio (normal text) | MUST   | WCAG 2.1 AA: minimum 4.5:1 against background              |
| Contrast ratio (large text)  | MUST   | WCAG 2.1 AA: minimum 3:1 for 18px+ or 14px+ bold           |
| Text overflow                | MUST   | No text beyond container boundary                          |
| Line length                  | SHOULD | Body text $\leq$ 80 characters per line                    |
| Line height                  | SHOULD | Body text at least 1.5 line-height                         |
| Scotopic mode                | MUST   | Minimum 4.5:1 contrast against covert background (#0a0c10) |
| Focus indicators             | MUST   | Visible focus indicators on all interactive elements       |

## 2. Audience Calibration

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DNS Tool serves three primary audiences with different reading contexts:

### The Engineer

**Context:** Analyzing a domain at a terminal, possibly at 2 AM, under incident pressure.

**Needs:** Precision, RFC references, verification commands, machine-parseable output.

**Vision concern:** Extended reading sessions on dark backgrounds. Font size and contrast directly affect error rate.

### The Executive

**Context:** Reading a brief before a board meeting or vendor assessment.

**Needs:** Risk posture in business terms, remediation cost/complexity, decision points.

**Vision concern:** May be reading on phone, tablet, or printed page. Layout must be responsive.

### The Older Professional

**Context:** 20+ years in infosec, may have age-related vision changes (presbyopia, reduced contrast sensitivity).

**Needs:** Everything both audiences need, plus accommodation for reduced visual acuity.

**Vision concern:** This is the primary reason the font-size floor exists. 10px text that “looks fine” to a 25-year-old developer may be unreadable to the 55-year-old CISO.

## 3. Enforcement

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### Automated Checks

| CHECK | TOOL | THRESHOLD |
|-------|------|-----------|
|-------|------|-----------|

|                          |                         |                                                         |
|--------------------------|-------------------------|---------------------------------------------------------|
| Font size floor          | CSS audit               | Zero sub-0.75rem values                                 |
| Contrast ratio           | squirreelscan           | Zero errors                                             |
| Text overflow            | Manual viewport testing | No horizontal scrollbar at 375px, 768px, 1024px, 1440px |
| Heading hierarchy        | squirreelscan           | No skipped levels                                       |
| Lighthouse accessibility | Chrome Lighthouse       | Score $\geq$ 95                                         |

## Manual Review Checklist

- CLARITY** Can a non-technical executive understand the main finding within 10 seconds?
- CLARITY** Does every acronym have a first-use expansion or link?
- CLARITY** Does every security finding include a “what to do” action?
- VISION** At 50cm / 20”, can all text be read on a 13” laptop at 100% zoom?
- VISION** In scotopic mode, does all text pass WCAG AA contrast (4.5:1)?
- VISION** On a 375px mobile viewport, does all content fit without horizontal scroll?
- VISION** Are interactive elements large enough for touch targets (44×44px)?

## 4. CSS Reference Values

Enforced values as of v26.46.12:

| PROPERTY                                            | VALUE                                        | CONTRAST RATIO | STANDARD                                                      |
|-----------------------------------------------------|----------------------------------------------|----------------|---------------------------------------------------------------|
| Font size floor                                     | 0.75rem (12px)                               | —              | All <code>.u-fs-*</code> utility classes enforce this minimum |
| <code>.text-muted</code>                            | <code>#9ca3af</code> on <code>#0d1117</code> | ~7.45:1        | WCAG AA (requires 4.5:1)                                      |
| <code>.card.bg-dark</code> <code>.text-muted</code> | <code>#adb5bd</code> on <code>#21262d</code> | ~8.8:1         | WCAG AA                                                       |
| Secondary text minimum                              | <code>#8b949e</code> on <code>#0d1117</code> | ~6.15:1        | Replaces prior <code>#6c757d</code> (~4.04:1)                 |

## 5. Connection to Project Philosophy

**[ACCOUNTABILITY]** — If the text is unreadable, the finding is unaccountable: it exists but cannot be acted upon.

**[ZERO\_WASTE]** — Text that cannot be read is wasted: it consumes space without delivering value.

**[VERBOSE+++++]** — Clarity does not mean brevity. It means every word earns its place by advancing the reader’s understanding.

**Vision as Ethics** — Causing eye strain to a reader who came for security intelligence is a design failure with real consequences.

### OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER

|                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                 |
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### KLEIN FOUR-GROUP $V_4$ — CLOSURE IN $O(2)$

$I \cdot \sigma_v \cdot C_2 \cdot \sigma_h \mid \sigma \circ \sigma = I \cdot C_2 = \sigma_v \circ \sigma_h \cdot$  Every element is its own inverse

“A score without a citation is an opinion. A finding without readability is invisible.”

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**Related documents:** [Founder’s Manifesto](#) · [Philosophical Foundations](#) · [Methodology Paper](#) · [Reference Library](#) · [Sources & Citations](#)